Getting Started

a) Retrieve your secret key from your Regfyl portal. **Note, this is only available on the primary account holder's dashboard.**

b) The secret key should be added as an API Key in Authorization. Add as a header (x-api-key).

c) Call the getCompany endpoint to extract the values of **companyName** and **rcNumber**. **Use this anywhere you see these parameters.**

d) In all API requests, you must pass the required parameters in the payload. Non-required parameters can be left undeclared.

e) Self-help with payload validation can be found here.

f) The API library is at Library.regfyl.com

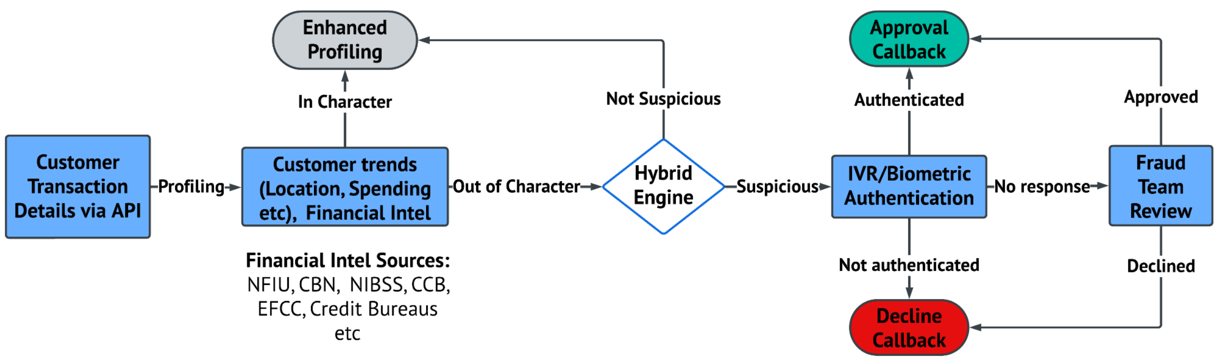
Transaction Monitoring Implementation Guide

Understanding the Transaction Monitoring Engine

The Transaction Monitoring Engine (TME) operates in progressive phases to detect high-risk transaction behaviors. During the initial phase, new customers are onboarded with limited historical data, and the system relies heavily on predefined, user-set rules to monitor transactions. Early transactions receive a generic "Data successfully added" response due to the lack of historical context. As transactions are processed, the system begins building customer profiles through data accumulation, allowing for more nuanced monitoring. High-risk behaviors are flagged based on these user-defined rules, though these flags do not necessarily indicate fraud. As the system evolves and gathers more data, it identifies potentially suspicious activities with greater accuracy, providing real-time API responses when sufficient context is available.

Note, advanced rule processing, such as evaluating multi-transaction scenarios, introduces slight delays in processing. For example, the system may flag a customer completing more than three transactions within five minutes, typically responding within 20 seconds after the threshold is crossed. Users are encouraged to implement latency periods, often aligned to high-risk rule processing times, to improve the system's effectiveness. While early monitoring depends on predefined rules, TME's hybrid approach continues to evaluate transactions based on these rules while leveraging accumulated data to provide context-aware alerts, ensuring comprehensive oversight of transaction activities.

The overall flow of the transaction monitoring is outlined below.



Transactions are passed via the post transaction endpoint. Regfyl then profiles the transaction for that customer using indicators such as their historical pending patterns, locations and other external/environmental data. For transactions that are not out-of-character, the details passed via api are added to enhance the customer's profile. However, the transactions that are out-of-character are then passed through our transaction monitoring engine to assess if it's suspicious. If suspicious, the transaction is flagged and then subsequently enters the authentication workflow. The authentication workflow is two fold.

1. **Automated authentication**: The customer is automatically contacted to authenticate the transaction using IVR/Biometrics. This feature is optional and must be manually requested.
2. **Manual review**: The fraud team reviews all flagged transactions including those that remain unresolved from (1) above. This is available via the dashboard.

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Simplifying Compliance

Transaction monitoring requires passing the customer's transactions to Regfyl via the post transaction endpoint. The full payload for transaction monitoring is available at the /postTransaction endpoint on library.regfyl.com however, the minimum mandatory payload for MFBs to successfully generate a valid XML is as below:

Key	**Description**
companyName	The name of the company that the transaction is associated with using this API.
rcNumber	The registration number of the company that the transaction is associated with using this API.
customerID	The customer's unique ID that you uniquely specify to identify the customer on your platform.
transactionAmount	Numerical value of the transaction e.g. 100
currency	Specify currency code of the transaction e.g. NGN for Naira, USD for US Dollars, GBP for Great Britain Pounds etc. The full list of currency codes is available via API here or downloaded here
transactionStatus	Set the status of the transaction as SUCCESSFUL, PENDING, REVERSAL, REFUND, DUPLICATE or DECLINED
transactionChannel	Set as CASH, CARD, VIRTUAL_WALLET, ATM, VIRTUAL_CARD, ONLINE or SHORT_CODE
authenticated	Set as YES if the transaction was authenticated using any of Password/PIN, Two-Factor Authentication (2FA), Biometric authentication, Out-of-Band Verification or any other layer of authentication. Set as NO

customerType	Set as INDIVIDUAL for individual customers or CORPORATE for corporate customers
transactionType	Set as INFLOW for inbound payments or OUTFLOW for outbound payments
environment	Set as SANDBOX for testing or PRODUCTION for live data
customerName	Full name of the customer
customerDOB	Date of birth of the customer in YYYY-MM-DD format
customerGender	Gender of the customer. Set as M or F
customerNationality	List of nationalities available here or here
idNumber	An official identification number for the customer. e.g. 12345
idType	The type of ID for the idNumber specified. Set as PASSPORT, DRIVERS_LICENSE, BVN, NIN, NATIONAL_ID, COMPANY_NUMBER, OTHER_ID
customerIdIssueDate	Issue date of the ID in YYYY-MM-DD format
phoneNo	Phone number of the customer in country code and number format e.g. 2348010000000
callbackURL	Callback url on your server to receive feedback for transactions sent for authentication Note: the x-Signature in the header is **hash_hmac('sha256', payload, secret_key)**
customerOccupation	Occupation of the customer e.g. Employed
customerFI	The recipient's financial institution code available here or here
customerFIType	Set as BANK or NON_BANK
customerAccountNumber	Account number of the recipient if available

customerSince	Date the customer opened an account in YYYY-MM-DD format e.g. 2024-01-01
transactionLocation	Location or city the transaction was completed e.g. Lagos
transactionCountry	Sender's country in the 2-code format e.g. NG for Nigeria. Full list available here or here
destinationCountry	Recipient's country in the 2-code format e.g. NG for Nigeria. List available here or here
purpose	Short summary of the reason for the transaction.
sourceFunds	Short description of source of funds e.g. Cash, customer's account etc
recipientName	Full name of the recipient
recipientAccountNumber	Account number of the recipient
recipientFI	The recipient's financial institution code available here or here
recipientFIName	Name of the recipient's financial institution
recipientFIType	Set as BANK or NON_BANK
collectionAccount	Set as YES or NO if the recipient account is a collection account
transactionReference	Unique reference number to identify the transaction generated by the client

API Responses

These are the real-time API responses:

Outcome	API response

API is successful and Transaction is NOT suspicious	{
	"description": "Data successfully added",
	"status": "200",
	"likelyFraud": "",
	"fraudReason": "",
	"predictiveConfidence": "",
	"customerID": "12345",
	"customerName": "John Doe",
	"reference": "",
	"fraudAction": "NONE",
	"transactionInvolvesHighRiskCountry": "Yes"
	}

API is successful but the transaction is suspicious.
 	
- predictiveConfidence is returned as LOW, MEDIUM or HIGH
- fraudAction is returned as either NONE or PAUSE
	{
	"description": "Data successfully added",
	"status": "200",
	"likelyFraud": "Yes",
	"fraudReason": "Transaction is incredibly high and unusual for this customer",
	"predictiveConfidence": "HIGH",
	"customerID": "12345",
	"customerName": "John Doe",
	"reference": "MTMzMjEMtMDQtMTc=00"
	"transactionReference": "IDI141",
	"fraudAction": "NONE",
	"transactionInvolvesHighRiskCountry": "Yes"
	}

API is not successful	{
	"description": "Upload error",
	"status": "400"
	}

API is not successful and Bad keyword e.g. for transactionType, a keyword	{ "description": "Invalid key-value pair or keyword",

beyond INFLOW or OUTFLOW is used	"status": "400" }

API is not successful because wallet balance is Insufficient	{
	"description": "Insufficient wallet balance or environment key-pair",
	"status": "400"
	}

API is not successful because Mandatory field missing	{
	"description": "At least 1 mandatory key is missing",
	"status": "400"
	}

API is not successful because Invalid method is selected	{
	"description": "Invalid method",
	"status": "400"
	}

Callbacks - Based on Rules

When a transaction is flagged by a rule, Regfyl sends the following callbacks:

{
"reference": "MTMzMjEMtMDQtMTc=00",
"transactionReference": "IDI141",
“rulePriority”: "LOW",
“customerID”: "IDI123",
“checkType”: "TRANSACTION_MONITORING",
}

Note, the rulePriority can be **LOW**, **MEDIUM** or **HIGH**

Dashboard Review Callbacks

When a transaction is flagged and reviewed on the Regfyl portal, Regfyl sends the following callbacks when the case is updated by the compliance team.

The 5 possible payloads include:

Outcome	**API response**

Not yet reviewed
 	
Default option until the transaction is reviewed.
	{
	"reference":
	"MTMzMjEMtMDQtMTc=00",
	"transactionReference": "IDI141",
	"status": "Not yet reviewed",
	“rulePriority”: "LOW",
	}

Reviewed - Further action required
 	
Implies that investigation is ongoing. Transaction is not yet Approved
	{
	"reference":
	"MTMzMjEMtMDQtMTc=00",
	"transactionReference": "IDI141",
	"status": "Reviewed - Further action required",
	“rulePriority”: "LOW",
	}

Reviewed - No further action required
 	
Implies that the transaction has been cleared as not suspicious.
	{
	"reference":
	"MTMzMjEMtMDQtMTc=00",
	"transactionReference": "IDI141",
	"status": "Reviewed - No further action required",
	“rulePriority”: "LOW",
	}

Reviewed - Declined
 	
Transaction has been declined by the compliance team
	{
	"reference":
	"MTMzMjEMtMDQtMTc=00",
	"transactionReference": "IDI141",
	"status": "Reviewed - Declined",
	“rulePriority”: "LOW",
	}

Reviewed - Ready for filing	{

Transaction is confirmed as suspicious and should be declined.	"reference": "MTMzMjEMtMDQtMTc=00", "transactionReference": "IDI141", "status": "Reviewed - Ready for filing", “rulePriority”: "LOW", }

Transaction Monitoring & Fraud Detection FAQ

Why can’t fraud be detected in the first few transactions? The system lacks historical data for new customers, so it relies on predefined rules and provides generic “Data added successfully” responses initially.

How does fraud detection improve over time? As the system processes more transactions, it builds customer profiles, enabling real-time, nuanced monitoring and flagging of suspicious behavior.

What role do user-defined rules play? Users set rules to flag patterns of high-risk activity, customizing the monitoring process to their needs.

Example of a rule? Flagging more than three transactions in five minutes. The system flags the fourth transaction for breaking this rule.

Why is there latency in flagging transactions? Complex rules, like multi-transaction patterns, take time to evaluate, typically within seconds or a few minutes.

How to handle latency? Delay value transfers for around 20 seconds to allow high-risk rule evaluation before completing the transaction.

Key takeaway? Fraud detection improves with historical data and user-defined rules. Combine rules, latency, and accumulated history for effective monitoring.

Customer Screening Implementation Guide

The service has three possibilities:

1. **Customer Screening**
   This includes PEP Screening, Sanctions Screening and Adverse Media screening.

Key	**Description**
companyName	The name of the company that the transaction is associated with using this API.
rcNumber	The registration number of the company that the transaction is associated with using this API.
customerID	The customer's unique ID that you uniquely specify to identify the customer on your platform.
customerType	Set as INDIVIDUAL
customerName	Pass the full name of the customer e.g. John Doe
YOB	Year of birth of the customer in YYYY format
callbackURL	Callback url on which to receive updates on the case Note: the x-Signature in the header is hash_hmac('sha256', payload, secret_key)
gender	Gender of the customer. Set as M or F or leave as empty
recurringCheck	Set as NO for once off checks.
Environment	Set as SANDBOX or PRODUCTION

2. **Customer Screening + ID Verification (BVN/NIN)**
   In addition to payload in (1), the following parameters should be passed:

verifyID	To verify the customer's ID, set as YES
country	ID verification is only available in Nigeria. Set as NG
idType	Types of ID available in Nigeria include BVN, NIN, PHONE_NUMBER
idNumber	Input the ID number based on the selected ID type.

3. **Customer Screening + ID Verification + Address Verification**
   In addition to the payload in (2), the following is required:

verifyAddress	To verify the customer's address stated on their ID, set as YES. Note, address verification is only available in Nigeria.
whatsappNumber	Required for address verification. Format is 2348000000000
verificationDate	Date to conduct the address verification in YYYY-MM-DD format and must be at least current day

API Responses

Outcome	API response

API is successful	{
	"description": "Data added successfully",
	"status": "200"
	}

API is not successful	{
	"description": "Upload error",
	"status": "400"
	}

API is not successful and Bad keyword e.g. for environment, a keyword beyond SANDBOX or PRODUCTION is used	{
	"description": "Invalid key-value pair or keyword",
	"status": "400"
	}

API is not successful because wallet balance is Insufficient	{
	"description": "Insufficient wallet balance or environment key-pair",
	"status": "400"
	}

API is not successful because Mandatory field missing	{
	"description": "At least 1 mandatory key is missing",
	"status": "400"
	}

API is not successful because Invalid method is selected	{
	"description": "Invalid method",
	"status": "400"
	}

Dashboard Review Callbacks

When a flagged customer is reviewed on the platform, Regfyl sends the following callbacks. The possible payloads include:

Outcome	**API response**

PEP	{
	"reference": "MTMzMjEMtMDQtMTc=00",
	"customerID": "12345",
	"checkType": "PEP",
	"status": "Not yet reviewed"
	}
	
	Note, status can be:
	- Not yet reviewed
	- Reviewed - Further action required
	- Reviewed - Confirmed
	- Reviewed - No further action required

	{
	"reference": "",
	"customerID": "12345",
	"checkType": "PEP",
	"status": ""
	}

Adverse Media	{
	"reference": "MTMzMjEMtMDQtMTc=00",
	"customerID": "12345",
	"checkType": "ADVERSE_MEDIA",
	"status": "Not yet reviewed"
	}
	
	Note, status can be:
	- Not yet reviewed
	- Reviewed - Further action required
	- Reviewed - Confirmed
	- Reviewed - No further action required

	{ "reference": "", "customerID": "12345",

	"checkType": "ADVERSE_MEDIA", "status": "" }

Sanctions Screening	{
	"reference": "MTMzMjEMtMDQtMTc=00",
	"customerID": "12345",
	"checkType": "SANCTIONS",
	"status": "Not yet reviewed"
	}
	
	Note, status can be:
	- Not yet reviewed
	- Reviewed - Further action required
	- Reviewed - Confirmed
	- Reviewed - No further action required

	{
	"reference": "",
	"customerID": "12345",
	"checkType": "SANCTIONS",
	"status": "Not yet reviewed"
	}

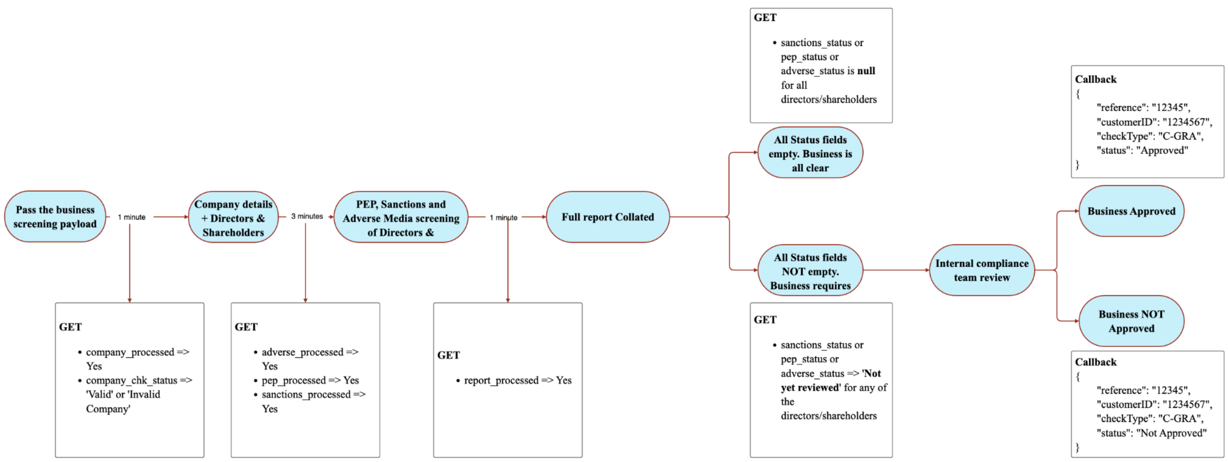
ID Verification	{ "reference": "MTMzMjEMtMDQtMTc=00", "customerID": "12345", "checkType": "ID_VERIFICATION", "status": "VERIFIED", "customerName": "John Doe", "idType": "BVN", "idNumber": "1234567", "country": "Nigeria", "dateOfBirth": "1900-01-01", "expiryDate": "1900-01-01", "address": "John Doe Street, Lagos", "phoneNo1": "08000000000", "phoneNo2": "08000000000", "user_email": "johndoe@email.com", "stateOfOrigin": "Kano", "gender": "M" or "F", "idFaceMatchOutcome": "MATCH" or "NO_MATCH"

	} Note, status can be: - Verified - Invalid or Unable to verify

Address Verification	{
	"reference": "MTMzMjEMtMDQtMTc=00",
	"customerID": "12345",
	"checkType": "ADDRESS_VERIFICATION",
	"status": "VERIFIED"
	}
	
	Note, status can be:
	- Verified
	- Not_verified
	- Unable to verify

Business Screening Flow Description

The Business Screening Flow provides a step-by-step overview of the automated screening process in Regfyl software, designed to assess companies and their key individuals (directors and shareholders) for regulatory compliance. This flow, outlined below, ensures a thorough risk assessment before onboarding new business entities, enhancing compliance and minimizing potential risks.



1. **Initiate Screening Process**: The process begins with passing the business screening payload, which initiates the compliance check. This action triggers an initial check on the company's basic details and key stakeholders. The endpoint is /postRiskAssessment

companyName	The name of the company that the transaction is associated with using this API.
rcNumber	The registration number of the company that the transaction is associated with using this API.
customerID	The customer's unique ID that you uniquely specify to identify the customer on your platform.

formType	Nature of business screening/risk assessment to be generated. Set as C-GRA
environment	Set as SANDBOX for testing or PRODUCTION for live data
registrationCountry	Registration jurisdiction of the entity. e.g. NG for Nigeria. List available here
registrationNumber	Registration number of the entity
addressVerification	Request for address verification. Must be set as Yes or No
monitoringFrequency	Request for address verification as part of C-GRA. Must be set as once-off or biannually or quarterly
callbackURL	Webhook url to notify you of the completion of the final outcome the risk assessment/business screening following review by the compliance team e.g. https://regfyl.com/notifications Note: the X-Signature in the header is hash_hmac('sha256', payload, secret_key)

2. **Retrieve Company Details and Key Individuals (1 Minute)**: Regfyl retrieves core company details, including information on directors and shareholders. A GET request (/RiskAssessmentDetails endpoint) returns the processing status (company_processed) and validity (company_chk_status) of the company.

3. **Screening for Risk Factors (3 Minutes)**: Regfyl screens up to 8 Directors and shareholders for PEP (Politically Exposed Persons), Sanctions, and Adverse Media records. A GET request returns the status

of the compliance checks (adverse_processed, pep_processed, sanctions_processed).

4. **Report Compilation (1 Minute)**: After screening, a full compliance report is collated. The report_processed status confirms completion, consolidating the findings of the screening steps.
5. **Decision-Making**: Based on the screening results, one of two paths is followed:
    - All Status Fields Empty: If no risk factors are found (fields for sanctions, PEP, and adverse media are clear), the business is marked "all clear" and approved.
    - Status Fields NOT Empty: If any risk factors are detected, the business requires further review by the internal compliance team.
6. **Internal Compliance Review**: In cases where risk factors are identified, the compliance team reviews the flagged issues in detail. They then determine if the business can proceed (approved) or should be denied based on regulatory risks.
7. **Callback Notification**: After the final decision, Regfyl sends a callback notification containing the business status. This includes a reference ID, customer ID, check type, and final status (Approved or Not Approved), confirming the business's compliance outcome.

Outcome	**API response**

Approved
 	
Implies that compliance has reviewed the business and cleared it for onboarding
	{
	  "reference": "MTMzMjEMtMDQtMTc=00",
	  "customerID": "12345",
	  "checkType": "C-GRA",
	  "status": "Approved"
	}

Not Approved
 	
Implies that compliance has reviewed the business has declined to onboard the company
	{
	  "reference": "MTMzMjEMtMDQtMTc=00",
	  "customerID": "12345",
	  "checkType": "C-GRA",

	"status": "Not Approved" }